Accuracy, Calibration and Script Robustness of Decision Models on Bengali: A Zero-Shot Benchmark with Hand-Written Romanization

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Abstract

System One decision models answer a typed question in a single forward pass and return a probability for each option, so a system can act on the confidence rather than on generated text. We evaluate one proprietary model (Jev) and eight open ones (five vLLM-SR d3 sizes, Laya-multilingual, laya-bangla, Julia-1, and Gemma-4-E4B scored by option probabilities) zero-shot on five Bengali tasks, in Bengali script and in romanized form. Jev has the highest score on all five tasks among the models run on the full test sets (Belebele 0.918, 95% CI 0.900–0.936), and the open d3-flash is level with it or above it on three of five on a subset of about 405 items per task. Synthetic romanization overstates how much accuracy a model loses on real Banglish: on 877 items that the author rewrote by hand, Jev loses 19.4 points under rule-based romanization but 11.2 under hand-written Banglish, a difference of +8.2 points (95% CI +5.1 to +11.1). The same ordering of models holds under both (Kendall $\tau = 0.89$ over 9 models), so synthetic romanization still ranks models, but it does not size their loss. Every open model is overconfident on Bengali; one temperature fitted on a disjoint development split removes most of the excess (Julia-1 on XNLI-bn: ECE 0.420 to 0.098), and removes less of it on romanized text. Code, metrics and the hand-written Banglish are released at `https://github.com/ihpolash/system-one-testing-for-bengali`.

1 Introduction

Zero-shot decision models are small encoders or distilled models that read a typed question and return one probability per option in a single pass. Their appeal is operational: the output is a calibrated-looking number, so a pipeline can accept confident answers and escalate the rest to a larger model or a person [4, 16]. Almost all evidence about them is in English. Chinese-Jev shows that language coverage cannot be assumed: Laya-multilingual reaches only 40.2% on Chinese tasks where Jev reaches 68.4% [18]. We found no study of a decision model on Bengali beyond the per-language Belebele and SIB-200 numbers released with the Jev benchmark [4], and none that asks whether a decision model's confidence still carries information in Bengali.

Bengali raises a second problem that English benchmarks do not. Much informal Bengali is typed in Latin letters ("Banglish") with no standard spelling. Studies of romanization in other Indic languages and in Bangla dialects build the romanized test text with a transliteration tool, and report large accuracy losses [11, 12]. A tool applies one spelling rule per letter; a person does not. Whether tool-made romanization is a fair stand-in for real Banglish has, to our knowledge, not been measured against hand-written text on model accuracy.

Study	Decision models	Bengali	Romanized	Calibration	Hand-written text
Deußer et al. [4]	Jev only	per-language file	Belebele, SIB-200	yes	no
Wang et al. [18]	trained model	no	no	yes	no
Khullar et al. [12]	no	no	tool-made	routing only	no
Jawad et al. [11]	no	dialects	tool-made	no	no
This paper	9 models	5 tasks	rule, natural-style, hand	yes	877 items

Table 1: What each closest study covers, from reading the papers. A "no" means the paper does not report it.

We make four contributions.

1. A zero-shot benchmark of nine decision models on five Bengali tasks in three text conditions (Bengali script, rule-based romanization, natural-style romanization), with cluster-bootstrap intervals, calibration, risk–coverage and option-rotation analyses (Section 4).
2. A set of 935 Bengali sentences (877 scorable items) that the author corrected and rewrote by hand in Banglish, and a paired comparison against two synthetic romanizers on the same items (Section 4.2).
3. Evidence that synthetic romanization overstates the loss by +8.2 points for Jev and by 8.2 to 17.1 points for the eight models above chance, while leaving the model ranking intact.
4. A harness that reproduces numbers published by independent parties for Jev, Laya-multilingual and laya-bangla (Section 4.5), which we use as a check on every result here.

We do not claim that this is the first Bengali evaluation of any decision model, and we report no result for Kev, d3-27B or Microsoft-Decision-1.

2 Related work

Table 1 lists what the closest prior studies cover. Decision-model evaluations in English cover calibration [16], annotation tasks with open models [9], option-name effects [17] and option-order stability [13]. Language-specific work is thinner: Chinese-Jev trains a Chinese model [18], and one study compares Jev's cultural-value answers in English and Arabic [2]. Script robustness has been studied for generative models only [11, 12]; the Bengali romanization corpus of Gharami et al. [7] generates its Bengali half by rule rather than collecting typed text. Calibration and selective prediction follow the standard measures [6, 8], and option-order sensitivity follows Zheng et al. [19]. The datasets we use are Belebele [3], SIB-200 [1], MASSIVE [5], SentNoB [10] and the Bengali XNLI translation.

3 Benchmark

Models. Jev (`jev-1.13.0`) is queried through its API. The open models run locally on a Kaggle GPU: Laya-multilingual (an mmBERT-based encoder [15]), laya-bangla v2 (a Bangla fine-tune of it, commit `4c5eb48932`), Julia-1 (144M parameters), and the vLLM-SR Decision 3.0 models d3-edge, d3-lite, d3-nano, d3-mini and d3-flash [14]. d3-flash (9B) ran with 4-bit NF4 quantization and on a hash-ordered subset of about 405 items per task, because of GPU time. We add Gemma-4-E4B-it

Dataset	Task	Options	Metric	Text origin	Test items (all / used)
Belebele	reading comprehension	4 options	accuracy	human translation (FLORES-200)	900 / 900
SIB-200	topic	7 options	accuracy	human translation (FLORES-200)	204 / 204
XNLI-bn	natural language inference	3 options	accuracy	machine translation of XNLI	4,895 / 1,500
MASSIVE bn-BD	intent	59 options	accuracy	human translation	2,974 / 1,500
SentNoB	sentiment	3 options	macro-F1	native social-media comments	1,586 / 1,586

Table 2: Tasks. XNLI-bn is the machine-translated Bengali XNLI, so its text is translationese; SentNoB is the only native Bengali set. The SentNoB split is a third-party mirror whose provenance we could not verify.

as a non-decision baseline: we score each option by the next-token probability of its label and normalize over the options. Kev and d3-27B were not run. Microsoft-Decision-1 is available only through an API and was not run.

Tasks. Table 2 lists the five Bengali tasks. Each item is posed as a typed single-choice question with English instructions and option names; only the passage or utterance changes with the condition. Large test sets are cut to a hash-ordered subset of 1,500 items so that smaller runs are nested in larger ones; a content-keyed response cache makes identical requests share one stored answer. The Bengali option names of Jev change accuracy by at most 0.8 points on SentNoB, XNLI-bn and SIB-200, so we keep English names. Accuracy is the headline metric, and macro-F1 for SentNoB.

Text conditions. *Bengali* is the dataset text. *Rule-based romanization* maps each Bengali letter to Latin letters with a fixed rule set and no diacritics. *Natural-style romanization* replaces each word by its most frequent typed spelling in a public corpus of natural Banglish (nahidstaq/bangla-transliteration-data) and falls back to rules tuned on held-out word types from that corpus. Both are synthetic. Belebele passages were not romanized.

Hand-written Banglish. The author took 935 Bengali sentences drawn from SentNoB, MASSIVE and XNLI-bn, corrected spelling errors in 81 of them, and wrote each sentence again in Banglish as they would type it. [TK: one sentence from the author: how the files were written, and whether to mention the archived tool-assisted draft of file 3 that was discarded] Items are matched to the datasets by the *original* text. XNLI-bn pairs need both sentences, so the set holds 877 scorable items: 494 MASSIVE, 325 SentNoB and 58 XNLI-bn pairs. Spelling correction alone moves Jev by +0.2 points (95% CI -0.2 to +0.7) and no model by more than 0.6 points, so we compare Banglish with the *corrected* Bengali, and apply both romanizers to the corrected text as well.

Statistics. Intervals are 95% percentile bootstraps over premise or passage clusters (Belebele, XNLI-bn) or items, and over items for the paired comparisons on the hand-written set. Calibration is expected calibration error with 15 equal-width bins (ECE), the signed gap between confidence and accuracy, Brier score and risk–coverage curves. Temperature scaling is fitted on a development split disjoint from the test items and applied unchanged to the test items; it uses native text only. Option-order sensitivity uses three cyclic rotations of the option list. We report no significance tests; comparisons are point estimates with intervals, and we do not correct for the many comparisons.

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918	0.897 (0.725)	0.751 (0.572)	0.757 (0.481)	0.623 (0.527)
Gemma-4-E4B	0.773	0.853 (0.730)	0.631 (0.519)	0.689 (0.429)	0.590 (0.533)
d3-mini	0.727	0.838 (0.549)	0.729 (0.562)	0.749 (0.383)	0.579 (0.486)
d3-nano	0.702	0.848 (0.377)	0.746 (0.497)	0.705 (0.221)	0.592 (0.408)
d3-lite	0.577	0.760 (0.265)	0.704 (0.463)	0.624 (0.121)	0.535 (0.343)
d3-edge	0.409	0.475 (0.235)	0.523 (0.445)	0.243 (0.044)	0.472 (0.366)
laya-bangla	0.313	0.750 (0.250)	0.625 (0.435)	0.575 (0.077)	0.464 (0.216)
Laya-multilingual	0.287	0.588 (0.275)	0.635 (0.383)	0.179 (0.045)	0.263 (0.141)
Julia-1	0.287	0.520 (0.284)	0.382 (0.316)	–	0.403 (0.299)
Majority class / chance	0.25	0.250	0.342	0.071	0.412

Table 3: Accuracy on Bengali script, and in brackets on rule-based romanization, rotation 0 (SentNoB: macro-F1). Belebele passages were not romanized. Julia-1 cannot take the 59 options of MASSIVE. Intervals are in Table 7.

On the d3-flash subset	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.840	0.897 (0.725)	0.752 (0.557)	0.799 (0.519)	0.637 (0.577)
d3-flash (4-bit)	0.730	0.892 (0.627)	0.828 (0.598)	0.819 (0.412)	0.653 (0.519)

Table 4: d3-flash (4-bit) and Jev on the same hash-ordered subset (204 to 405 items per task); romanized in brackets. †4-bit quantization of the 9B checkpoint.

4 Results

4.1 Accuracy on Bengali script and romanized text

Table 3 and Figure 1 give the main results. Jev has the highest point estimate on all five tasks among the models run on the full sets; the closest open models are d3-nano on XNLI-bn (0.746 against 0.751) and d3-mini (0.729), and the intervals in Table 7 overlap. On its subset, d3-flash reaches 0.828 on XNLI-bn against Jev's 0.752, and trails on Belebele (Table 4); it is above Jev on three tasks, but the subset holds only 204 to 405 items per task, so we treat it as level. Gemma-4-E4B is the most robust open model to romanization: it keeps 0.730 of 0.853 on SIB-200, where d3-mini keeps 0.549 of 0.838. laya-bangla improves on its base model on SIB-200 (0.750 against 0.588) and MASSIVE (0.575 against 0.179), but stays below Jev, d3-mini and Gemma-4-E4B on every task. On SentNoB, Laya-multilingual (0.263) and Julia-1 (0.403) are below the majority-class level (0.412), and Julia-1 cannot take the 59-option MASSIVE menu at all. Laya-multilingual's SentNoB macro-F1 depends on wording: 0.263 with the default instruction, 0.321 with option names only and 0.376 with Bengali wording.

4.2 Hand-written Banglish

Figure 2 and Table 5 compare four spellings of the same 877 items. Every model above chance scores higher on hand-written Banglish than on rule-based romanization, by 8.2 to 17.1 points; for Jev the difference is +8.2 (95% CI +5.1 to +11.1). Natural-style romanization is closer to the hand-written text but still harder for every model, so tuning a romanizer on natural typing narrows the gap without closing it. The gain depends on the task (Figure 3): for Jev it is +13.6 points on MASSIVE (+9.7 to +17.4), +8.6 on XNLI-bn pairs (+0.0 to +17.2; 58 pairs) and +0.0 on SentNoB (-4.9 to +4.6). We read the MASSIVE gain as an effect of short, imperative utterances with many

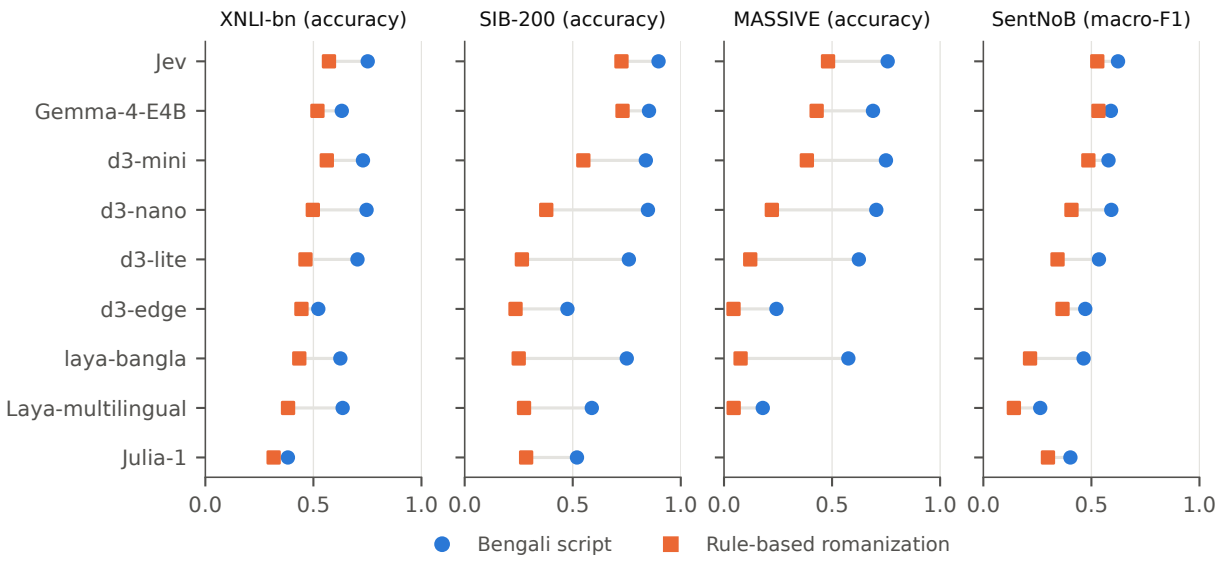


Figure 1: Bengali script against rule-based romanization for each model and task. Lines join the two conditions for the same model.

common English loan words, which people spell consistently; we did not test that. Julia-1, near chance on these items, shows no gain.

The ordering of models is nearly the same under all three Latin-script conditions (Kendall $\tau = 0.89$ for rule-based and 0.94 for natural-style against hand-written text, over 9 models). The eight models are scored on the same items, so this is one comparison, not eight independent ones.

4.3 Calibration and selective prediction

Models differ widely in how overconfident they are. Raw ECE on Bengali script (mean over XNLI-bn, MASSIVE and SentNoB) ranges from 0.074 for d3-mini to 0.432 for Julia-1 (Table 6). A single temperature fitted on a disjoint Bengali development split cuts ECE for every model on Bengali text, for example Julia-1 on XNLI-bn from 0.420 to 0.098. On rule-based romanization the same temperature leaves a higher ECE for eight of the nine models (d3-mini is the exception): Jev's scaled ECE is 0.095 against 0.047 on Bengali, and laya-bangla's is 0.172 against 0.034. Figure 4 shows the reliability curves, and Figure 5 the risk–coverage curves: confidence still ranks answers within each condition, but romanization lowers accuracy at every coverage level, including among the most confident answers.

A confidence gate set on native development text to reach 90% accuracy does not transfer. On MASSIVE, Jev accepts 69% of Bengali items at 91% accuracy and 23% of romanized items at 88%; d3-mini accepts 68% at 90% and 22% at 83%. For laya-bangla the accepted romanized items are 17% accurate, at 1% coverage. So a gate tuned on Bengali keeps its accuracy for Jev and d3-mini on this task, but only by refusing most romanized input, and fails for laya-bangla. These rows rest on a single task and, for several models, on few accepted items; we treat them as exploratory.

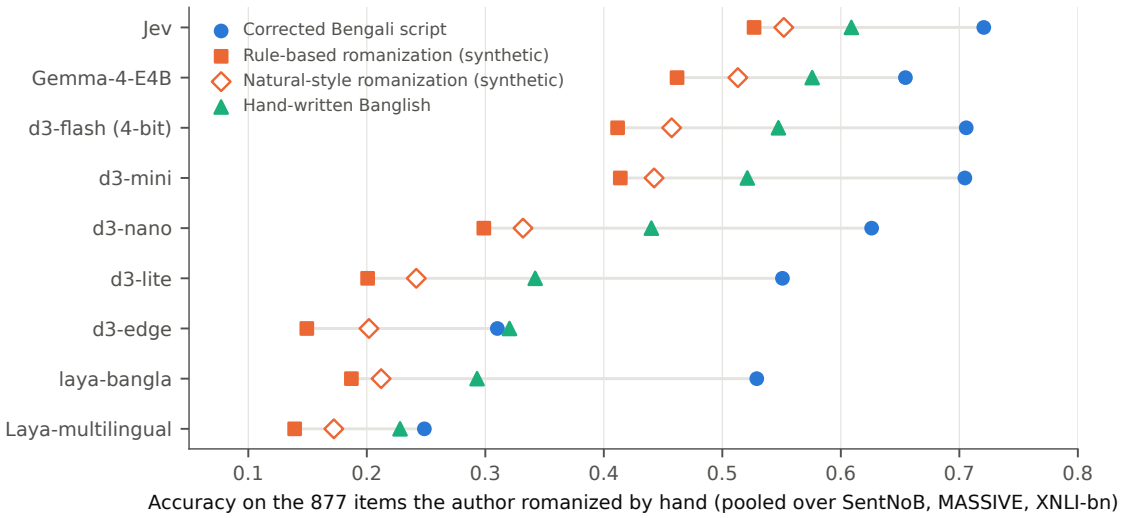


Figure 2: Accuracy of each model on the same 877 items in four spellings.

	Accuracy				Change in points (95% CI)	
Model	Bengali	Rule-based	Natural-style	Banglish	Banglish − Bengali	Banglish − rule-based
Jev	0.721	0.527	0.552	0.609	-11.2 [-14.1, -8.2]	+8.2 [+5.1, +11.1]
d3-flash (4-bit)†	0.706	0.412	0.457	0.547	-15.8 [-19.3, -12.7]	+13.6 [+10.7, +16.8]
Gemma-4-E4B	0.655	0.462	0.513	0.576	-7.9 [-10.7, -5.1]	+11.4 [+8.8, +14.3]
d3-mini	0.705	0.414	0.442	0.521	-18.4 [-22.0, -14.9]	+10.7 [+8.0, +13.6]
d3-nano	0.626	0.299	0.332	0.440	-18.6 [-22.4, -15.1]	+14.1 [+11.3, +17.4]
d3-lite	0.551	0.201	0.242	0.342	-20.9 [-24.2, -17.1]	+14.1 [+11.3, +17.2]
d3-edge	0.310	0.149	0.202	0.320	+1.0 [-2.7, +4.7]	+17.1 [+14.1, +19.7]
laya-bangla	0.529	0.187	0.212	0.293	-23.6 [-27.3, -20.0]	+10.6 [+8.0, +13.3]
Laya-multilingual	0.249	0.139	0.172	0.228	-2.1 [-5.0, +0.9]	+8.9 [+6.6, +11.2]
Julia-1‡	0.428	0.441	0.418	0.384	-4.4 [-11.0, +2.1]	-5.7 [-11.2, +0.0]

Table 5: Accuracy on the 877 hand-written items (pooled over MASSIVE, SentNoB and XNLI-bn pairs; accuracy for all three) and paired differences with 95% bootstrap intervals over items. †d3-flash is 4-bit. ‡Julia-1 answers only the 383 items it can take.

4.4 Option-order sensitivity

Under three rotations of the options, Jev gives the same answer 93% of the time and d3-nano 89%, d3-lite 87% and Gemma-4-E4B 81%. The encoder models are far less stable: laya-bangla 67%, Laya-multilingual 53% and Julia-1 35% (Figure 6; per-task values in Table 8).

4.5 Checks on the harness

Recomputed from the Jev authors' released responses, Belebele ben_Beng accuracy is 0.9167; our own run gives 0.918. For Laya-multilingual we obtain MASSIVE 0.179 and SIB-200 0.588, against 0.18 and 0.57 to 0.58 reported by the laya-bangla authors; for laya-bangla, SIB-200 0.750 against 0.76. Resending 300 Jev requests gave the same answer 99% of the time, and a different Laya package version changed no answer on 6,563 requests. We could not compare laya-bangla's MASSIVE score (0.575 against 0.81 reported) because the authors use a two-step scenario-then-intent protocol and we ask one 59-way question. The study used 92,618 Jev requests, about 50.9M input

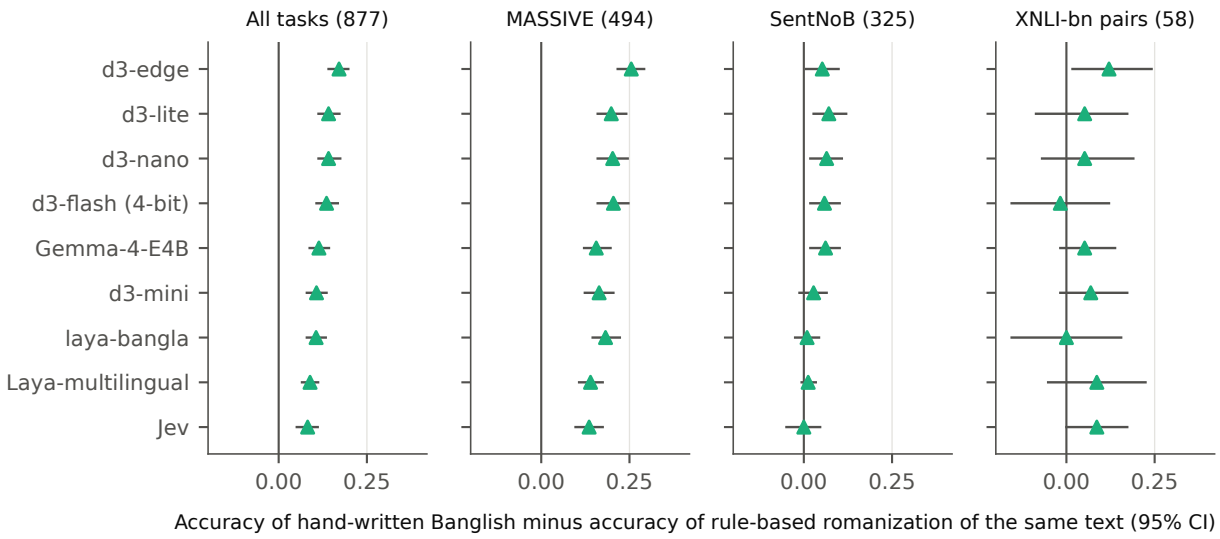


Figure 3: Accuracy of hand-written Banglish minus rule-based romanization of the same text, pooled and by task. Bars are 95% intervals.

	Bengali script		Rule-based romanization	
Model	ECE raw	ECE scaled	ECE raw	ECE scaled
Jev	0.126	0.047	0.146	0.095
Gemma-4-E4B	0.303	0.073	0.391	0.082
d3-mini	0.074	0.036	0.079	0.035
d3-nano	0.083	0.041	0.122	0.054
d3-lite	0.076	0.040	0.172	0.108
d3-edge	0.174	0.044	0.202	0.085
laya-bangla	0.204	0.034	0.331	0.172
Laya-multilingual	0.347	0.080	0.530	0.209
Julia-1	0.432	0.104	0.494	0.154

Table 6: ECE (15 bins) on Bengali script and rule-based romanization, before and after one temperature fitted on a native-text development split. Mean over XNLI-bn, MASSIVE and SentNoB.

tokens, an input-token cost of about $2.14 at $0.042 per million tokens, and about 26 GPU hours on Kaggle. Latency was measured in mixed conditions and is not comparable between models; we do not report it.

5 Discussion

What the synthetic number does and does not tell you. A romanization tool is a good way to rank models and a poor way to size their loss. Written by a person, Banglish keeps more of the information a model needs than a tool's output does: common words are spelled the way models have seen them, and spelling errors in the source text are not copied across. We did not test which of these matters; the corrected-text control shows only that spelling errors in the original Bengali are not the cause of the gap. A practitioner who planned for a 19.4-point loss on Banglish input from tool-made test data would plan for about 8 points too many for Jev.

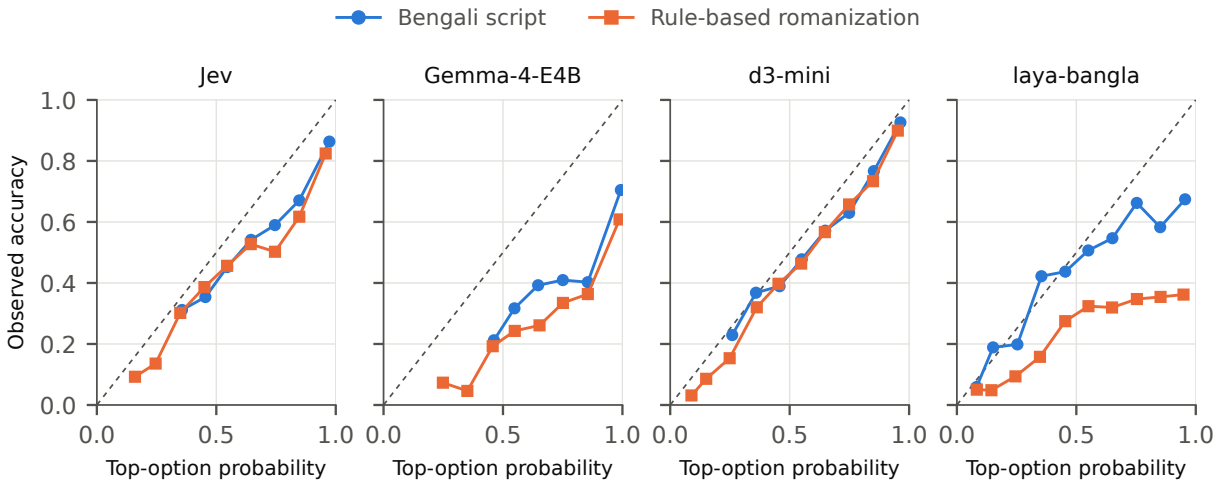


Figure 4: Reliability of four models on MASSIVE, XNLI-bn and SentNoB pooled. The dashed line is perfect calibration.

Choosing a model. For Bengali script, the open d3-mini, d3-nano and Gemma-4-E4B come within a few points of Jev on several tasks, so a decision model's advantage here is not in accuracy alone. Gemma-4-E4B is the most robust open model to romanization but is also among the least calibrated before scaling (ECE 0.303), which temperature scaling mostly removes. The encoder models need care: their answers change with option order, and Laya-multilingual's SentNoB score depends on instruction wording.

6 Limitations

One person wrote all of the Banglish, so the gap we report is for that typing style; a second typist on a subset would show how much it varies. MASSIVE supplies 56% of the hand-written items and XNLI-bn only 58 pairs, and XNLI-bn and MASSIVE are translations, so only SentNoB is native Bengali text. On SentNoB the Banglish is not easier than rule-based romanization, and we do not know whether that is a property of social-media text or of our sample. d3-flash ran 4-bit on a subset, d3-27B and Kev were not run, and Julia-1 cannot take MASSIVE. Training-data contamination is unknown for Jev, Gemma and the Qwen-based d3 models; laya-bangla was trained on MASSIVE and SIB-200 training splits. The option names are English, and the Bengali option names we tried were written by the author and not checked by a second native speaker. Calibration and gate results are exploratory, rest on few accepted items for some models, and use no multiplicity correction. We use one prompt format per model family; a different format may change the ranking, as Laya's wording sensitivity shows.

7 Conclusion

On five Bengali tasks, Jev leads, several small open models come close on Bengali script, and every open model needs a recalibration that transfers only in part to romanized text. For evaluating Banglish input, tool-made romanization gives the right ordering of models but too large a loss; testing on hand-written text is needed to size it.

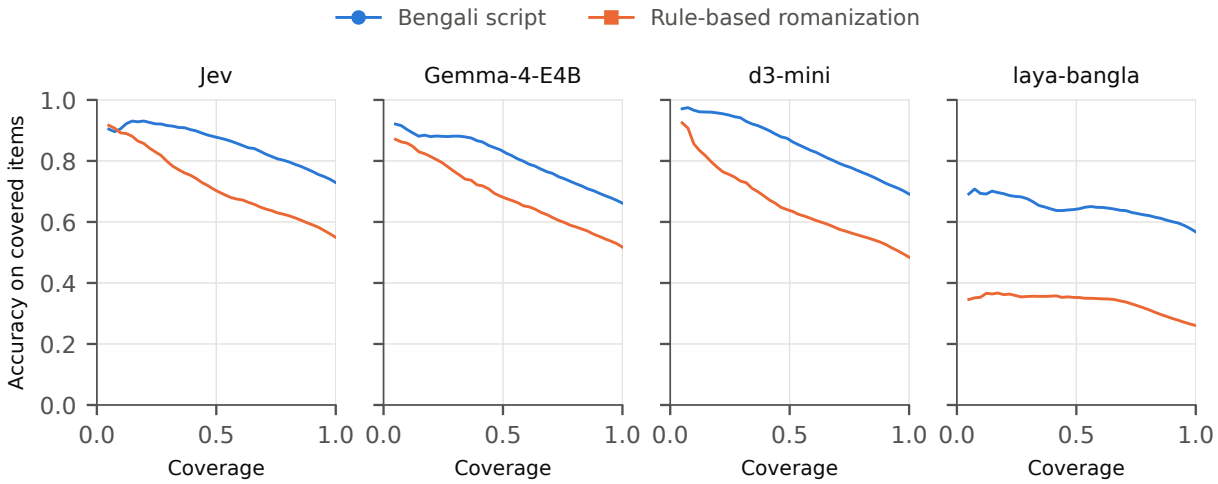


Figure 5: Accuracy on the covered items as coverage grows from the most confident answers.

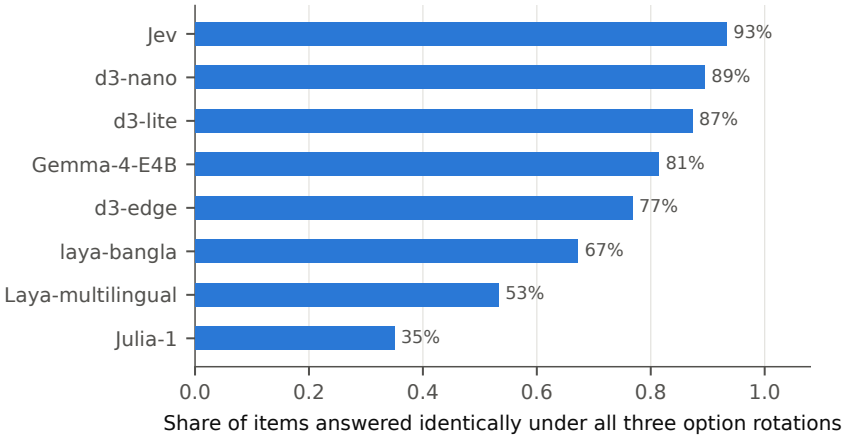


Figure 6: Share of items answered identically under all three option rotations, over all tasks.

Availability. The repository is at `https://github.com/ihpolash/system-one-testing-for-bengali`. It holds the harness, the analysis code that generates every table and number in this paper, the summary result tables, and the hand-written Banglish for the MASSIVE and XNLI-bn sentences. SentNoB is distributed under a no-derivatives licence, so the repository does not contain its text or the Banglish of its sentences; the item-level predictions and Jev's raw responses (provider terms) are also not included.

AI assistance. An AI assistant (Claude, Anthropic) was used to write and run evaluation code, to run the experiments, to generate tables and figures from the stored results, and to draft and edit the text. The author designed the study, chose the datasets and models, corrected the Bengali text and wrote all of the Banglish by hand, checked the results against independent published numbers, and reviewed every claim. The author is solely responsible for the content. The assistant is not an author.

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A Additional tables

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.918 [0.900, 0.936]	0.897 [0.858, 0.936]	0.751 [0.729, 0.772]	0.757 [0.735, 0.778]	0.623 [0.603, 0.646]
Gemma-4-E4B	0.773 [0.746, 0.801]	0.853 [0.804, 0.902]	0.631 [0.608, 0.655]	0.689 [0.664, 0.713]	0.590 [0.569, 0.614]
d3-mini	0.727 [0.697, 0.757]	0.838 [0.789, 0.887]	0.729 [0.707, 0.751]	0.749 [0.727, 0.769]	0.579 [0.556, 0.604]
d3-nano	0.702 [0.672, 0.731]	0.848 [0.799, 0.892]	0.746 [0.724, 0.768]	0.705 [0.682, 0.729]	0.592 [0.566, 0.615]
d3-lite	0.577 [0.545, 0.609]	0.760 [0.701, 0.819]	0.704 [0.680, 0.727]	0.624 [0.602, 0.647]	0.535 [0.511, 0.558]
d3-edge	0.409 [0.375, 0.438]	0.475 [0.407, 0.544]	0.523 [0.499, 0.547]	0.243 [0.221, 0.264]	0.472 [0.447, 0.497]
laya-bangla	0.313 [0.283, 0.344]	0.750 [0.691, 0.809]	0.625 [0.598, 0.651]	0.575 [0.549, 0.599]	0.464 [0.440, 0.490]
Laya-multilingual	0.287 [0.259, 0.316]	0.588 [0.524, 0.652]	0.635 [0.608, 0.660]	0.179 [0.159, 0.198]	0.263 [0.245, 0.284]
Julia-1	0.287 [0.257, 0.317]	0.520 [0.456, 0.588]	0.382 [0.359, 0.406]	–	0.403 [0.378, 0.425]

Table 7: Accuracy on Bengali script with 95% cluster-bootstrap intervals (SentNoB: macro-F1).

Model	Belebele	SIB-200	XNLI-bn	MASSIVE	SentNoB
Jev	0.9 / 93%	1.0 / 97%	0.5 / 97%	1.7 / 89%	0.9 / 95%
Gemma-4-E4B	1.4 / 74%	0.5 / 93%	3.3 / 86%	4.3 / 70%	0.1 / 94%
d3-nano	1.1 / 82%	0.5 / 96%	0.7 / 96%	1.1 / 85%	0.4 / 95%
d3-lite	1.2 / 82%	1.0 / 96%	0.7 / 96%	0.5 / 80%	1.2 / 92%
d3-edge	0.3 / 75%	4.4 / 83%	1.9 / 92%	3.3 / 49%	1.4 / 90%
laya-bangla	2.1 / 28%	2.0 / 94%	1.2 / 89%	0.3 / 66%	2.8 / 88%
Laya-multilingual	1.1 / 15%	1.0 / 84%	0.1 / 91%	8.7 / 29%	4.6 / 80%
Julia-1	1.8 / 31%	2.9 / 74%	1.9 / 24%	–	10.8 / 45%

Table 8: Spread of accuracy across the three option rotations (points) / share of items answered identically under all three.